

Adeel Ahmad

Staff Platform Engineer | Agentic AI, Full-Stack & AI Security on AWS | AWS SA Professional

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SUMMARY

Staff-level engineer and architect with 19 years of hands-on experience building software, cloud platforms and distributed systems, including more than ten years on AWS. At Commonwealth Bank's AI Centre of Excellence, I turn ambiguous GenAI and data problems into secure production systems spanning agentic applications, lineage graphs, identity, evaluation, DevSecOps and AI security. Previously an AWS consultant embedded with enterprise clients and Founder/CTO of a 40-person consultancy; still hands-on in Python, TypeScript, Go and Java.

EXPERIENCE

Commonwealth Bank of Australia · Melbourne

Mar 2025 – Present

Staff Platform Engineer (MLOps), AI Centre of Excellence · AI Acceleration

- Served as the technical anchor for CommDataIQ from early design through production, covering backend architecture, authentication, deployment readiness, production debugging and staging-to-production transitions across software, data and infrastructure teams.
- Worked backwards from data-steward needs and narrowed a bank-wide lineage brief to three priority journeys; with AWS engineers, delivered a security-reviewed prototype in three weeks, an ROI model and a six-month roadmap. Later demonstrated an enriched Amazon Neptune lineage graph with 113,000 nodes and 146,000 edges.
- Designed a hybrid lineage-extraction approach: deterministic Python parsers and ASTs for known Control-M, JCL and SQL patterns, with schema-constrained LLM fallback for edge cases. Users can ask lineage questions in plain English while retaining traceability to source metadata and code.
- Established workload-identity and OAuth 2.0 patterns for service-to-service and agent-to-tool access; led Ping integration, risk/security assessments, penetration-test scoping, Project Purple remediation and evidence preparation for APRA and internal risk reviews.
- Raised the squad's DevSecOps maturity score from 1.81 to 2.75 in one cycle (target 2.3) by tying 14 controls to delivery OKRs; introduced versioned prompts, LLM-judge evaluation, SAST/DAST/SCA, staged deployment and rollback evidence.
- Ran authorised prompt-injection testing of an internal multi-agent assistant, demonstrated a tool-boundary weakness and triggered guardrail work; also contributed to the design of an AI observability and control-plane initiative spanning three teams.
- Mentored engineers through design decisions, debugging and PR reviews, helping them reason independently; ran brownbags on authentication and ECS Service Connect and coordinated technical direction with cross-functional stakeholders.

LyreBird Health · Melbourne

Aug 2024 – Oct 2024

Tech Lead, AWS and GenAI · AI medical scribe

- Built the AWS foundation for patient health data: Control Tower multi-account setup, SCPs, least-privilege IAM, KMS rotation, encryption in transit and at rest, and GitHub Actions CI/CD using OIDC and Secrets Manager.
- Replaced free-form clinical note generation with two-pass structured extraction in which each statement cited its supporting transcript lines; added prompt versioning and LLM-as-judge evaluation across Amazon Bedrock and OpenAI models.

Cevo (AWS Consulting Partner) · Melbourne

Feb 2022 – Jul 2024

AWS Cloud Consultant · embedded with enterprise clients; multiple performance awards

- Led government-mandated Consumer Data Right REST APIs at a major Australian energy provider, covering the solution-architect role early, leading two teams (8 engineers) and building the AWS data platform; production release cleared about \$2.5M of regulatory exposure.
- Resolved a production outage the same day on Amazon Neptune and Gremlin after inheriting an unfamiliar system; the bounded-query fix became the pattern across all endpoints. Separately found accounts with up to 48,000 transactions beyond a 10,000-result limit and drove a paginated design.
- In June 2023, pitched the client CTO on disposable browser-based cloud development environments where an AI agent could take a Jira story, return a plan and produce a pull request for review; also proposed a self-hosted coding assistant to keep source code in-house.
- After 1.5 weeks of discovery for a meat exporter, built a serverless prototype overnight using SES, S3, Step Functions, Textract, Comprehend and Bedrock. The next-morning demo changed the CTO's direction; the production design was about \$700/year for work performed manually by roughly 40 people.
- Delivered production retrieval over a 290TB legacy SharePoint estate using Amazon Q Business.

- Founded and grew the consultancy to about 40 people across two countries; led sales from discovery through proposals and pricing, and delivered 80+ web, cloud and digital-preservation engagements inside client teams at Yale, Harvard, Indiana University/CPB, VIAA, the UN and Intel.
- Designed Aviary on AWS for Yale's Fortunoff Archive (4,400+ Holocaust testimonies with strict access controls); built InsureSign from scratch on Java/Spring, S3 and SQS before Formstack acquired it.
- Co-built an open-source archival system with Indiana University/CPB covering 2.5M records and 1M+ hours of media, and Junkipedia, a misinformation-monitoring platform adopted by the UN; wrote the company engineering playbook covering AWS, MFA, secure coding, standards and onboarding.
- Moved development into browser-based VS Code environments after power failures repeatedly cost engineers work. The platform began on one European server shared by 12 developers, then evolved toward Firecracker and Cloud Hypervisor microVMs with template-based workspaces and idle shutdown.

Earlier: PureLogics — Senior Software Engineer, then Co-Partner and CTO (2009–2012), where our Facebook application won FBfund. APTLOGIX — Software Developer on e-commerce and C#.NET (2007–2009).

SELECTED OPEN SOURCE & RESEARCH

- **AgentRC (agentrc.ai).** Open specification and Go CLI for packaging AI agents. An Agentfile declares identity, capabilities, SOPs and policy, builds into OCI labels and can be enforced with Cedar; reference implementation includes a BuildKit frontend.
- **agentic-agile + agent-handoff.** Claude Code workflow with planning, test-first development, review and final verification across isolated git worktrees; agent-handoff is a pure-Go tool for carrying settled decisions, open questions and task history between agent sessions.
- **Early agentic systems.** Built an AutoGen notebook in Oct 2023 modelling an AWS Professional Services-style team of GPT-4 agents that scopes an architecture brief, checks it against the Well-Architected Framework and drafts specialist-agent code after human approval. Presented a serverless, local-model "second brain" in 2024.
- **MLX reasoning research.** Created mlx-grpo-trainer and mlx-guided-grpo for GRPO and curriculum-guided reasoning training on Apple Silicon; current ESC-GRPO work uses masked worked-solution chunks, LoRA and multiple rollouts per prompt.
- **Systems and storage engineering.** Maintains a substantial JuiceFS fork with iSCSI, multi-tier cache, packaging and observability work; long-term hands-on background with Proxmox, Ceph, ZFS, networking and private-cloud infrastructure.
- **Full-stack and security research.** Built React/Next.js/Tauri applications with TypeScript, Prisma, PostgreSQL and SQLite, including MCP assistants and an agent-design studio. Reported unauthorised model selection and system-message tampering in OpenAI GPTs through Bugcrowd.

SKILLS

Languages & full-stack: Python, TypeScript/JavaScript, Go, Java, Ruby, PHP; React, Next.js, Tauri, Node.js, Spring and Rails.

GenAI & agents: Amazon Bedrock, Bedrock AgentCore, Amazon Q Business, MCP, AutoGen, LangChain; RAG, reranking, embeddings, prompt/version evaluation, LLM-as-judge, guardrails, MLX, LoRA and GRPO.

AWS & platforms: Lambda, ECS, EKS, S3, RDS, IAM, KMS, Secrets Manager, CloudWatch, X-Ray, Control Tower; CloudFormation, Terraform, Pulumi, GitHub Actions, Kubernetes, Docker, Firecracker and Proxmox.

Data, identity & security: PostgreSQL, MongoDB, Redis, Neptune/Gremlin, OpenSearch; OAuth 2.0, OIDC, JWT, SAML, workload identity, Cedar, SAST/DAST/SCA, prompt-injection testing, APRA, HIPAA and CDR.

Systems foundations: Proxmox, QEMU/KVM, Firecracker, Ceph, ZFS, NFS/iSCSI, OpenVSwitch, VXLAN, MikroTik, Cisco, WireGuard and Zero Trust networking.

Customer delivery: Discovery workshops, working backwards, architecture trade-offs, rapid prototyping, executive demos, proposals, mentoring and production incident response.

RECOGNITION

Top of CBA's Cyber Champions leaderboard in February 2026; team placed third in CBA's GenAI hackathon; multiple Cevo performance awards; GitHub Arctic Code Vault Contributor.

CERTIFICATIONS & EDUCATION

AWS Certified Solutions Architect – Professional (valid to Jul 2027; AWS-certified since 2016)

BS Computer Science, Institute of Management Sciences, Lahore (2007–2011)